CBSE Exam Preparation

Class 10 Science

Ch 1: Chemical Reactions and Equations

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41 one-mark, 24 short-answer and 7 long-answer questions with answers

Based on the official NCERT textbook (ncert.nic.in). Prepared 11 November 2026.

Chapter 1: Chemical Reactions and Equations

Source: https://ncert.nic.in/textbook/pdf/jesc101.pdf

1. Summary Notes

Chemical change and chemical reaction

- Everyday examples of chemical change: milk left at room temperature in summer, iron tawa/pan/nail left in humid air (rusting), fermentation of grapes, cooking of food, digestion of food and respiration.
- In all of these, the nature and identity of the starting substance changes. Whenever a chemical change occurs, a chemical reaction has taken place.
- Activity 1.1: a clean magnesium ribbon burns in air with a dazzling white flame and changes into a white powder, magnesium oxide (MgO).
- Activity 1.2: lead nitrate solution + potassium iodide solution gives a yellow precipitate of lead iodide.
- Activity 1.3: zinc granules + dilute HCl or H_2SO_4 give bubbles of hydrogen gas, and the flask becomes warm.
- A chemical reaction can be recognised by any of these observations: change in state, change in colour, evolution of a gas, change in temperature.

Chemical equations

- A word-equation writes a reaction in short form: Magnesium + Oxygen → Magnesium oxide.
- Reactants are the substances that undergo chemical change; they are written on the left-hand side (LHS). Products are the new substances formed; they are written on the right-hand side (RHS).
- A plus sign (+) separates two reactants or two products. The arrow points towards the products and shows the direction of the reaction.
- A chemical equation uses chemical formulae instead of words, e.g. $Mg + O_2 \rightarrow MgO$.
- If the number of atoms of each element is not the same on both sides, the equation is unbalanced and is called a skeletal chemical equation.

Balanced chemical equations

- Law of conservation of mass: mass can neither be created nor destroyed in a chemical reaction. So the number of atoms of each element must be the same before and after the reaction.
- Example of a balanced equation: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$ (Zn 1, H 2, S 1, O 4 on both sides).
- Steps to balance $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ (hit-and-trial method):
- Step I: draw boxes around each formula; never change anything inside a box, i.e. never change a formula (write $4H_2O$, not H_2O_4 or $(H_2O)_4$).
- Step II: list the number of atoms of each element on both sides.
- Step III: start with the compound having the maximum number of atoms (Fe_3O_4) and the element with most atoms in it (oxygen): $Fe + 4H_2O \rightarrow Fe_3O_4 + H_2$.
- Step IV: balance hydrogen: $Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$.
- Step V: balance the remaining element, iron: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$.
- Step VI: count atoms on both sides again to check that the equation is balanced.
- Step VII: add physical-state symbols: (s) solid, (l) liquid, (g) gas, (aq) aqueous solution in water. $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$. Here (g) with water shows steam is used.
- Reaction conditions such as temperature, pressure or catalyst are written above/below the arrow, e.g. $CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$ at 340 atm; $6CO_2(aq) + 12H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 6O_2(aq) + 6H_2O(l)$ in the presence of sunlight and chlorophyll (photosynthesis).

Combination reaction

- A reaction in which a single product is formed from two or more reactants.
- Activity 1.4: quick lime reacts vigorously with water to form slaked lime, releasing a large amount of heat: $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$.
- Whitewashing: a solution of slaked lime reacts slowly with CO_2 in air to form a thin layer of calcium carbonate in 2-3 days, giving a shiny finish: $Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$. Marble is also $CaCO_3$.
- Other examples: burning of coal, $C(s) + O_2(g) \rightarrow CO_2(g)$; formation of water, $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$.

Exothermic and endothermic reactions

- Exothermic reactions release heat along with the formation of products.
- Examples: burning of natural gas, $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$; respiration, $C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l) + energy$; decomposition of vegetable matter into compost.
- Respiration is exothermic: carbohydrates in food (rice, potatoes, bread) are broken down into glucose, which combines with oxygen in our cells to give energy.
- Endothermic reactions absorb energy (heat, light or electricity). Decomposition reactions are generally endothermic because energy is needed to break the reactant down.

Decomposition reaction

- A reaction in which a single reactant breaks down to give simpler products. It is the opposite of a combination reaction.
- Thermal decomposition (by heat): Activity 1.5 - green ferrous sulphate crystals lose water and change colour on heating, and a smell of burning sulphur is noticed: $2FeSO_4(s) \rightarrow Fe_2O_3(s) + SO_2(g) + SO_3(g)$ (on heating).
- Limestone decomposes on heating: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$. Quick lime (CaO) is used in the manufacture of cement.
- Activity 1.6 - heating lead nitrate gives brown fumes of nitrogen dioxide: $2Pb(NO_3)_2(s) \rightarrow 2PbO(s) + 4NO_2(g) + O_2(g)$.
- Electrolytic decomposition (by electricity): Activity 1.7 - electrolysis of water, $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$. The volume of gas collected at the cathode (hydrogen) is double the volume at the anode (oxygen). Hydrogen burns with a pop sound; oxygen relights a glowing splinter.
- Photolytic decomposition (by light): Activity 1.8 - white silver chloride turns grey in sunlight: $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$. Silver bromide behaves similarly: $2AgBr(s) \rightarrow 2Ag(s) + Br_2(g)$. These reactions are used in black and white photography.

Displacement reaction

- A more reactive element displaces a less reactive element from its compound.
- Activity 1.9: iron nails dipped in copper sulphate solution become brownish and the blue colour of the solution fades: $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$.
- Other examples: $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$; $Pb(s) + CuCl_2(aq) \rightarrow PbCl_2(aq) + Cu(s)$.
- Zinc and lead are more reactive than copper, so they displace copper from its compounds.

Double displacement reaction and precipitation

- A reaction in which there is an exchange of ions between the reactants.
- Activity 1.10: sodium sulphate solution + barium chloride solution gives a white insoluble substance, barium sulphate: $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$.
- An insoluble substance formed in a reaction is called a precipitate. Any reaction that produces a precipitate is a precipitation reaction.
- Lead nitrate + potassium iodide (Activity 1.2) is also a double displacement reaction giving a yellow precipitate of lead iodide: $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$.

Oxidation and reduction (redox reactions)

- Activity 1.11: heating copper powder in air forms a black coating of copper(II) oxide: $2Cu + O_2 \rightarrow 2CuO$ (on heating).
- Passing hydrogen over hot CuO turns the black coating brown again as copper is obtained: $CuO + H_2 \rightarrow Cu + H_2O$ (on heating).
- Oxidation: a substance gains oxygen or loses hydrogen. Reduction: a substance loses oxygen or gains hydrogen.
- In $CuO + H_2 \rightarrow Cu + H_2O$, copper(II) oxide is reduced to copper and hydrogen is oxidised to water.
- Reactions in which one reactant is oxidised and the other is reduced are called oxidation-reduction or redox reactions.
- Other examples: $ZnO + C \rightarrow Zn + CO$ (ZnO reduced, C oxidised); $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$ (MnO_2 reduced, HCl oxidised).

Effects of oxidation in everyday life

- Corrosion: when a metal is attacked by substances around it such as moisture and acids, it corrodes. Examples: reddish-brown coating on iron (rusting), black coating on silver, green coating on copper.
- Corrosion causes damage to car bodies, bridges, iron railings, ships and all objects made of metals, especially iron. Huge amounts of money are spent every year to replace damaged iron.
- Rancidity: when fats and oils are oxidised, they become rancid and their smell and taste change.
- Rancidity is prevented by adding antioxidants to foods containing fats and oils, and by storing food in air-tight containers. Chips packets are flushed with an inert gas such as nitrogen to prevent oxidation.

2. Key Points to Remember

- ★ Signs of a chemical reaction: change in state, change in colour, evolution of a gas, change in temperature.
- ★ Law of conservation of mass: mass can neither be created nor destroyed in a chemical reaction; atoms of each element are equal on both sides of a balanced equation.
- ★ Skeletal equation = unbalanced chemical equation. Balancing method used in NCERT = hit-and-trial method.
- ★ Never change the formula of a substance while balancing; change only the coefficients.
- ★ State symbols: (s) solid, (l) liquid, (g) gas, (aq) aqueous.
- ★ $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$
- ★ Combination: $A + B \rightarrow AB$. $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$ (quick lime → slaked lime).
- ★ Whitewash: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ (shiny finish after 2-3 days; marble is $CaCO_3$).
- ★ Decomposition: $AB \rightarrow A + B$; types - thermal (heat), electrolytic (electricity), photolytic (light).
- ★ $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$ (green crystals change colour; smell of burning sulphur).
- ★ $CaCO_3 \rightarrow CaO + CO_2$ (quick lime used for cement).
- ★ $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$ (brown fumes of NO_2).
- ★ Electrolysis of water: $2H_2O \rightarrow 2H_2 + O_2$; hydrogen volume is twice that of oxygen.
- ★ $2AgCl \rightarrow 2Ag + Cl_2$ and $2AgBr \rightarrow 2Ag + Br_2$ in sunlight; used in black and white photography.
- ★ Displacement: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (blue solution fades, nail turns brown).

★ Double displacement: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow$ (white ppt) + 2NaCl; $Pb(NO_3)_2 + 2KI \rightarrow PbI_2\downarrow$ (yellow ppt) + $2KNO_3$.

★ Precipitate: insoluble solid formed in a reaction; such reactions are precipitation reactions.

★ Oxidation = gain of oxygen / loss of hydrogen. Reduction = loss of oxygen / gain of hydrogen.

★ Redox: $CuO + H_2 \rightarrow Cu + H_2O$; $ZnO + C \rightarrow Zn + CO$; $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$.

★ Exothermic reactions give out heat (respiration, burning of natural gas, composting); decomposition reactions are generally endothermic.

★ Corrosion: rust on iron (reddish brown), black coating on silver, green coating on copper.

★ Rancidity: oxidation of fats and oils; prevented by antioxidants, air-tight containers and nitrogen flushing.

3. One Mark Questions (1 mark each)

1. **Which of the statements about the reaction $2PbO(s) + C(s) \rightarrow 2Pb(s) + CO_2(g)$ are incorrect? Statement 1: Lead is getting reduced. Statement 2: Carbon dioxide is getting oxidised. Statement 3: Carbon is getting oxidised. Statement 4: Lead oxide is getting reduced.** [MCQ]
   (a) 1 and 2
   (b) 1 and 3
   (c) 1, 2 and 3
   (d) all
   Ans. (a) 1 and 2. Lead is a product (it is PbO that gets reduced), and CO_2 is a product formed by the oxidation of carbon.

2. **$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$. This reaction is an example of a** [MCQ]
   (a) combination reaction
   (b) double displacement reaction
   (c) decomposition reaction
   (d) displacement reaction
   Ans. (d) displacement reaction

3. **What happens when dilute hydrochloric acid is added to iron fillings?** [MCQ]
   (a) Hydrogen gas and iron chloride are produced
   (b) Chlorine gas and iron hydroxide are produced
   (c) No reaction takes place
   (d) Iron salt and water are produced
   Ans. (a) Hydrogen gas and iron chloride are produced

4. **The colour of the precipitate formed when lead nitrate solution is mixed with potassium iodide solution is** [MCQ]
   (a) white
   (b) yellow
   (c) green
   (d) brown
   Ans. (b) yellow (lead iodide)

5. **When ferrous sulphate crystals are heated, the gases evolved are** [MCQ]
   (a) SO_2 only
   (b) SO_3 only
   (c) SO_2 and SO_3
   (d) O_2 and SO_2
   Ans. (c) SO_2 and SO_3

6. **In the electrolysis of water, the ratio of volumes of hydrogen and oxygen collected is** [MCQ]
   (a) 1 : 2
   (b) 2 : 1
   (c) 1 : 1
   (d) 1 : 8
   Ans. (b) 2 : 1

7. **Which of the following is a photolytic decomposition reaction?** [MCQ]
   (a) $CaCO_3 \rightarrow CaO + CO_2$
   (b) $2H_2O \rightarrow 2H_2 + O_2$
   (c) $2AgCl \rightarrow 2Ag + Cl_2$
   (d) $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
   Ans. (c) $2AgCl \rightarrow 2Ag + Cl_2$ (takes place in sunlight)

8. **The brown fumes evolved on heating lead nitrate are of** [MCQ]
   (a) nitrogen dioxide
   (b) oxygen
   (c) lead oxide
   (d) nitric oxide
   Ans. (a) nitrogen dioxide (NO_2)

9. **$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$ is an example of** [MCQ]
   (a) displacement reaction
   (b) double displacement reaction
   (c) combination reaction
   (d) decomposition reaction
   Ans. (b) double displacement reaction (also a precipitation reaction)

10. **In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, the substance oxidised is** [MCQ]
   (a) CuO
   (b) H_2
   (c) Cu
   (d) H_2O
   Ans. (b) H_2 (it gains oxygen to form water)

11. **Which of the following is NOT an exothermic process?** [MCQ]
   (a) Respiration
   (b) Burning of natural gas
   (c) Decomposition of vegetable matter into compost
   (d) Decomposition of calcium carbonate
   Ans. (d) Decomposition of calcium carbonate (it needs heat, so it is endothermic)

12. **Chips packets are flushed with which gas to prevent rancidity?** [MCQ]
   (a) Oxygen
   (b) Hydrogen
   (c) Nitrogen
   (d) Carbon dioxide
   Ans. (c) Nitrogen

13. **The colour of the coating formed on copper when it corrodes is** [MCQ]
   (a) black
   (b) reddish brown
   (c) green
   (d) white
   Ans. (c) green

14. **Substances that undergo chemical change in a reaction are called _____.** [Fill in the blank]
Ans. reactants

15. **An unbalanced chemical equation is called a _____ chemical equation.** [Fill in the blank]
Ans. skeletal

16. **The method of balancing used by making trials with the smallest whole-number coefficients is called the _____ method.** [Fill in the blank]
Ans. hit-and-trial

17. **The state symbol (aq) stands for _____.** [Fill in the blank]
Ans. aqueous (present as a solution in water)

18. **Calcium oxide is commonly called _____ and calcium hydroxide is called _____.** [Fill in the blank]
Ans. quick lime; slaked lime

19. **Magnesium ribbon burns with a dazzling _____ flame.** [Fill in the blank]
Ans. white

20. **An insoluble substance formed during a reaction is called a _____.** [Fill in the blank]
Ans. precipitate

21. **Decomposition of silver bromide in sunlight is used in _____ photography.** [Fill in the blank]
Ans. black and white

22. **Gain of hydrogen by a substance is called _____.** [Fill in the blank]
Ans. reduction

23. **Oxidation of fats and oils resulting in a change in their smell and taste is called _____.** [Fill in the blank]
Ans. rancidity

24. **Decomposition reactions are generally exothermic.** [True/False]
Ans. False. Decomposition reactions generally require energy (heat, light or electricity), so they are endothermic.

25. **While balancing an equation, the formula H_2O can be changed to H_2O_2 to balance oxygen.** [True/False]
Ans. False. Formulae must never be changed; only coefficients (e.g. $2H_2O$) can be changed.

26. **Zinc can displace copper from copper sulphate solution.** [True/False]
Ans. True. Zinc is more reactive than copper.

27. **Respiration is an endothermic process.** [True/False]
Ans. False. Respiration is an exothermic process; it releases energy.

28. **Hydrogen gas burns with a pop sound.** [True/False]
Ans. True.

29. **Why should a magnesium ribbon be cleaned before burning in air?** [Very short answer]
Ans. To remove the protective layer of magnesium oxide (and dirt) on its surface so that magnesium burns readily in air.

30. **Write the balanced equation for: Hydrogen + Chlorine → Hydrogen chloride.** [Very short answer]
Ans. $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

31. **What is the chemical formula of marble?** [Very short answer]
Ans. $CaCO_3$ (calcium carbonate)

32. Name the gas that relights a glowing splinter. [Very short answer]
Ans. Oxygen

33. What change in colour is observed when white silver chloride is left in sunlight? [Very short answer]
Ans. It turns grey, because silver chloride decomposes into silver and chlorine.

34. Why does the colour of copper sulphate solution change when an iron nail is dipped in it? [Very short answer]
Ans. Iron displaces copper from copper sulphate to form ferrous sulphate (light green), so the blue colour fades: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$.

35. Name the product formed when quick lime is used to make cement after heating limestone. [Very short answer]
Ans. Calcium oxide (quick lime), formed by $CaCO_3 \rightarrow CaO + CO_2$.

36. Why do we apply paint on iron articles? [Very short answer]
Ans. Paint prevents moisture and air from coming in contact with iron, so it prevents rusting (corrosion).

37. Oil and fat containing food items are flushed with nitrogen. Why? [Very short answer]
Ans. Nitrogen is an inert gas; it prevents oxidation of fats and oils, so the food does not become rancid.

38. Assertion (A): A chemical equation must be balanced. Reason (R): Mass can neither be created nor destroyed in a chemical reaction. [Assertion-Reason]
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true
Ans. (a) Both A and R are true and R is the correct explanation of A.

39. Assertion (A): The reaction $CaO + H_2O \rightarrow Ca(OH)_2$ is a combination reaction. Reason (R): Heat is released in this reaction. [Assertion-Reason]
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true
Ans. (b) Both A and R are true but R is not the correct explanation of A. It is a combination reaction because a single product is formed from two reactants.

40. Assertion (A): In the reaction $ZnO + C \rightarrow Zn + CO$, zinc oxide is reduced. Reason (R): Zinc oxide loses oxygen in this reaction. [Assertion-Reason]
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true
Ans. (a) Both A and R are true and R is the correct explanation of A.

41. Assertion (A): Silver articles become black after some time when exposed to air. Reason (R): Silver is oxidised by oxygen to form a white oxide. [Assertion-Reason]
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true
Ans. (c) A is true but R is false. Silver corrodes and gets a black coating; it is not a white oxide.

4. Short Answer Questions (2 / 3 marks)

1. **What is a balanced chemical equation? Why should chemical equations be balanced?** [2 marks]
   Ans.
   - A balanced chemical equation has an equal number of atoms of each element on both sides (reactants and products).
   - Equations must be balanced to satisfy the law of conservation of mass: mass can neither be created nor destroyed in a chemical reaction.

2. **Translate into a balanced chemical equation: Hydrogen sulphide gas burns in air to give water and sulphur dioxide.** [2 marks]
   Ans. $2H_2S(g) + 3O_2(g) \rightarrow 2H_2O(l) + 2SO_2(g)$

3. **Write the balanced chemical equations for: (i) Barium chloride + Aluminium sulphate → Barium sulphate + Aluminium chloride (ii) Sodium + Water → Sodium hydroxide + Hydrogen** [2 marks]
   Ans.
   - (i) $3BaCl_2 + Al_2(SO_4)_3 \rightarrow 3BaSO_4 + 2AlCl_3$
   - (ii) $2Na + 2H_2O \rightarrow 2NaOH + H_2$

4. **Write balanced chemical equations with state symbols: (i) Solutions of barium chloride and sodium sulphate in water react to give insoluble barium sulphate and a solution of sodium chloride. (ii) Sodium hydroxide solution reacts with hydrochloric acid solution to produce sodium chloride solution and water.** [2 marks]
   Ans.
   - (i) $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$
   - (ii) $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$

5. **A solution of a substance 'X' is used for whitewashing. (i) Name the substance 'X' and write its formula. (ii) Write the reaction of 'X' with water.** [2 marks]
   Ans.
   - (i) X is calcium oxide (quick lime), CaO.
   - (ii) $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq)$ + Heat. The calcium hydroxide (slaked lime) solution formed is used for whitewashing.

6. **Why is respiration considered an exothermic reaction? Explain.** [2 marks]
   Ans.
   - During digestion, carbohydrates are broken down into glucose.
   - Glucose combines with oxygen in our cells and releases energy: $C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l)$ + energy.
   - Since energy is released, respiration is an exothermic reaction.

7. **Why are decomposition reactions called the opposite of combination reactions? Write equations for these reactions.** [2 marks]
   Ans.
   - In a combination reaction, two or more reactants combine to form a single product, e.g. $CaO + H_2O \rightarrow Ca(OH)_2$.
   - In a decomposition reaction, a single reactant breaks down into two or more simpler products, e.g. $CaCO_3 \rightarrow CaO + CO_2$ (on heating).
   - So the two are opposite processes.

8. **Why does the colour of copper sulphate solution change when an iron nail is dipped in it?** [2 marks]

Ans.

- Iron is more reactive than copper, so it displaces copper from copper sulphate solution: $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$.
- Ferrous sulphate formed is light green, so the blue colour of the solution fades, and a brown coating of copper forms on the nail.

9. **Explain the following in terms of gain or loss of oxygen with two examples each: (a) Oxidation (b) Reduction** [2 marks]

Ans.

- (a) Oxidation is gain of oxygen. Examples: $2Cu + O_2 \rightarrow 2CuO$; in $CuO + H_2 \rightarrow Cu + H_2O$, H_2 is oxidised to H_2O.
- (b) Reduction is loss of oxygen. Examples: in $CuO + H_2 \rightarrow Cu + H_2O$, CuO is reduced to Cu; in $ZnO + C \rightarrow Zn + CO$, ZnO is reduced to Zn.

10. **A shiny brown coloured element 'X' on heating in air becomes black in colour. Name the element 'X' and the black coloured compound formed.** [2 marks]

Ans.

- X is copper (Cu); the black compound is copper(II) oxide (CuO).
- $2Cu + O_2 \rightarrow 2CuO$ (on heating)

11. **Explain the following terms with one example each: (a) Corrosion (b) Rancidity** [2 marks]

Ans.

- (a) Corrosion: a metal is attacked by substances around it such as moisture and acids and gets damaged. Example: rusting of iron (reddish-brown coating).
- (b) Rancidity: fats and oils get oxidised and their smell and taste change. Example: chips or butter left open for a long time taste and smell bad.

12. **What does one mean by exothermic and endothermic reactions? Give examples.** [2 marks]

Ans.

- Exothermic reactions release heat. Example: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) + \text{heat}$ (burning of natural gas).
- Endothermic reactions absorb energy. Example: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ on heating; $2AgCl \rightarrow 2Ag + Cl_2$ in sunlight.

13. **Why should a magnesium ribbon be cleaned before it is burnt in air? Write the balanced equation for the reaction and state what you observe.** [3 marks]

Ans.

- Magnesium ribbon has a layer of magnesium oxide on its surface which slows down burning; cleaning with sandpaper removes this layer so that magnesium burns readily.
- Equation: $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
- Observation: it burns with a dazzling white flame and changes into a white powder (magnesium oxide).

14. **Balance the equation $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ step by step and write it with state symbols.** [3 marks]

Ans.

- Oxygen: 1 on LHS, 4 on RHS, so write $4H_2O$: $Fe + 4H_2O \rightarrow Fe_3O_4 + H_2$.
- Hydrogen: 8 on LHS, 2 on RHS, so write $4H_2$: $Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$.
- Iron: 1 on LHS, 3 on RHS, so write 3Fe: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$.
- With state symbols: $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$ (water is used as steam).

15. **Write balanced chemical equations for the following and identify the type of each reaction: (a) Potassium bromide(aq) + Barium iodide(aq) → Potassium iodide(aq) + Barium bromide(s) (b) Zinc carbonate(s) → Zinc oxide(s) + Carbon dioxide(g) (c) Hydrogen(g) + Chlorine(g) → Hydrogen chloride(g) (d) Magnesium(s) + Hydrochloric acid(aq) → Magnesium chloride(aq) + Hydrogen(g)** [3 marks]

Ans.

- (a) $2KBr(aq) + BaI_2(aq) \rightarrow 2KI(aq) + BaBr_2(s)$: double displacement reaction
- (b) $ZnCO_3(s) \rightarrow ZnO(s) + CO_2(g)$: decomposition reaction
- (c) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$: combination reaction
- (d) $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$: displacement reaction

16. **Write the balanced chemical equation for the following and identify the type of reaction in each case: (a) Thermit reaction, iron(III) oxide reacts with aluminium and gives molten iron and aluminium oxide. (b) Magnesium ribbon is burnt in an atmosphere of nitrogen gas to form solid magnesium nitride. (c) Chlorine gas is passed in an aqueous potassium iodide solution to form potassium chloride solution and solid iodine.** [3 marks]

Ans.

- (a) $Fe_2O_3(s) + 2Al(s) \rightarrow Al_2O_3(s) + 2Fe(l)$: displacement reaction (also redox)
- (b) $3Mg(s) + N_2(g) \rightarrow Mg_3N_2(s)$: combination reaction
- (c) $Cl_2(g) + 2KI(aq) \rightarrow 2KCl(aq) + I_2(s)$: displacement reaction

17. **Write one equation each for decomposition reactions where energy is supplied in the form of heat, light and electricity.** [3 marks]

Ans.

- Heat (thermal decomposition): $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
- Light (photolytic decomposition): $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ in sunlight
- Electricity (electrolytic decomposition): $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$

18. **What is the difference between displacement and double displacement reactions? Write equations for these reactions.** [3 marks]

Ans.

- In a displacement reaction, a more reactive element displaces a less reactive element from its compound. Example: $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$.
- In a double displacement reaction, two compounds exchange their ions. Example: $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$.
- In displacement only one element is replaced; in double displacement both compounds exchange their parts.

19. **In the refining of silver, the recovery of silver from silver nitrate solution involves displacement by copper metal. Write down the reaction involved.** [3 marks]

Ans.

- Copper is more reactive than silver, so it displaces silver from silver nitrate solution.
- $Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$
- The solution turns blue due to copper nitrate and silver metal is deposited.

20. **What do you mean by a precipitation reaction? Explain by giving examples.** [3 marks]

Ans.

- A reaction that produces an insoluble substance (precipitate) is called a precipitation reaction.
- Example 1: $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$; a white precipitate of barium sulphate forms.
- Example 2: $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$; a yellow precipitate of lead iodide forms.
- Both are double displacement reactions.

21. **Identify the substances that are oxidised and the substances that are reduced: (i) $4Na(s) + O_2(g) \rightarrow 2Na_2O(s)$ (ii) $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$ (iii) $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$** [3 marks]

Ans.

- (i) Na is oxidised (gains oxygen); O_2 is reduced.
- (ii) H_2 is oxidised to H_2O; CuO is reduced to Cu.
- (iii) HCl is oxidised (loses hydrogen) to Cl_2; MnO_2 is reduced (loses oxygen) to $MnCl_2$.

22. **Why is silver chloride stored in dark coloured bottles? Explain with an equation.** [3 marks]

Ans.

- Silver chloride decomposes in the presence of sunlight: $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$.
- White silver chloride turns grey as silver is formed.
- Coloured (dark) bottles cut off light and prevent this photolytic decomposition.

23. **What is observed when lead nitrate is heated? Write the equation and name the type of reaction.** [3 marks]

Ans.

- Brown fumes of nitrogen dioxide are given off and a yellowish residue of lead oxide remains.
- $2Pb(NO_3)_2(s) \rightarrow 2PbO(s) + 4NO_2(g) + O_2(g)$ (on heating)
- It is a thermal decomposition reaction.

24. **What are the effects of corrosion? How can rancidity of food be prevented?** [3 marks]

Ans.

- Corrosion damages car bodies, bridges, iron railings, ships and all objects made of metals, especially iron; a huge amount of money is spent each year to replace damaged iron.
- Rancidity can be prevented by adding antioxidants to food containing fats and oils.
- Storing food in air-tight containers and flushing packets (e.g. chips) with nitrogen also slows oxidation.

5. Long Answer Questions (5 marks)

1. **What is a chemical equation? Explain the steps of balancing a chemical equation by the hit-and-trial method, taking the reaction of iron with steam as an example. Why do we write state symbols and reaction conditions?** [5 marks]

Ans. A chemical equation represents a chemical reaction using the formulae of reactants and products. Reactant: iron and steam; products: iron(II, III) oxide and hydrogen.

- Skeletal equation: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$.
- Step I: draw boxes around each formula; do not change anything inside the boxes.
- Step II: list atoms on each side: Fe 1 and 3, H 2 and 2, O 1 and 4.
- Step III: start with the compound having most atoms (Fe_3O_4) and its element with most atoms (O). Put 4 before H_2O: $Fe + 4H_2O \rightarrow Fe_3O_4 + H_2$.
- Step IV: balance H: 8 on LHS, so put 4 before H_2: $Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$.
- Step V: balance Fe: put 3 before Fe: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$.
- Step VI: check: Fe 3 = 3, H 8 = 8, O 4 = 4. The equation is balanced.
- Step VII: add state symbols: $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$.
- State symbols (s, l, g, aq) make the equation more informative by giving the physical state of each substance; conditions such as temperature, pressure, catalyst or sunlight are written above/below the arrow, e.g. $CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$ at 340 atm.

2. **What is a decomposition reaction? Describe with equations the three ways in which energy can be supplied for decomposition. Mention one use of each type.** [5 marks]
   Ans. A decomposition reaction is one in which a single reactant breaks down to give simpler products. These reactions generally need energy, so they are endothermic.
   - Thermal decomposition (heat): heating ferrous sulphate - green crystals change colour with a smell of burning sulphur: $2FeSO_4(s) \rightarrow Fe_2O_3(s) + SO_2(g) + SO_3(g)$. Heating limestone: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$; the quick lime formed is used in the manufacture of cement. Heating lead nitrate gives brown fumes: $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$.
   - Electrolytic decomposition (electricity): passing current through water: $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$. Hydrogen collects at the cathode and oxygen at the anode; the volume of hydrogen is double that of oxygen. Used to obtain hydrogen and oxygen.
   - Photolytic decomposition (light): white silver chloride turns grey in sunlight: $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$; similarly $2AgBr(s) \rightarrow 2Ag(s) + Br_2(g)$. Used in black and white photography.
3. **Describe an activity to show the electrolysis of water. Write the equation, state the observations and how you would test the gases produced.** [5 marks]
   Ans.
   - Take a plastic mug, drill two holes at its base and fit rubber stoppers in them. Insert carbon electrodes in the stoppers and connect them to a 6 V battery.
   - Fill the mug with water so that the electrodes are immersed, and add a few drops of dilute sulphuric acid.
   - Take two test tubes filled with water and invert them over the two carbon electrodes.
   - Switch on the current and leave the apparatus undisturbed for some time.
   - Observation: bubbles form at both electrodes and displace water in the test tubes. The volume of gas collected over the cathode (negative electrode) is double that over the anode.
   - Equation: $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$ (electric current).
   - Test: bring a burning candle close to the mouth of each tube. Hydrogen burns with a pop sound; oxygen makes a glowing splinter/candle burn brighter.
   - This is an electrolytic decomposition reaction.
4. **Explain displacement and double displacement reactions with activities and equations. How is a precipitation reaction related to them?** [5 marks]
   Ans. Displacement reaction:
   - A more reactive element displaces a less reactive element from its compound.
   - Activity: dip clean iron nails in copper sulphate solution for about 20 minutes. The nails become brownish and the blue colour of the solution fades: $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$.
   - Other examples: $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$; $Pb(s) + CuCl_2(aq) \rightarrow PbCl_2(aq) + Cu(s)$. Zinc and lead are more reactive than copper.

   Double displacement reaction:
   - There is an exchange of ions between the reactants.
   - Activity: mix sodium sulphate solution with barium chloride solution. A white insoluble substance, barium sulphate, forms: $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$.

   Precipitation reaction:
   - The insoluble substance formed is a precipitate, and any reaction that produces a precipitate is a precipitation reaction.
   - Most double displacement reactions in solution, like the one above and $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s)$ (yellow) $+ 2KNO_3(aq)$, are precipitation reactions.

5. **What are oxidation and reduction? Explain with an activity why oxidation and reduction occur together (redox reaction). Give two more examples.** [5 marks]
   Ans.
   - Oxidation: gain of oxygen or loss of hydrogen by a substance. Reduction: loss of oxygen or gain of hydrogen by a substance.
   - Activity: heat about 1 g copper powder in a china dish. The brown surface becomes coated with black copper(II) oxide because copper combines with oxygen: $2Cu + O_2 \rightarrow 2CuO$ (copper is oxidised).
   - Now pass hydrogen gas over the heated black material. The black coating turns brown again because CuO loses oxygen: $CuO + H_2 \rightarrow Cu + H_2O$.
   - In this reaction CuO is reduced to Cu and at the same time H_2 is oxidised to H_2O. One reactant gets oxidised while the other gets reduced, so it is called an oxidation-reduction or redox reaction.
   - Example 1: $ZnO + C \rightarrow Zn + CO$; ZnO is reduced and C is oxidised.
   - Example 2: $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$; MnO_2 is reduced and HCl is oxidised.

6. **What is a combination reaction? Explain with the reaction of quick lime and water, and describe how whitewashing gives a shiny finish. Is this reaction exothermic? Give other examples of exothermic reactions.** [5 marks]
   Ans.
   - A combination reaction is one in which a single product is formed from two or more reactants.
   - Activity: add water slowly to a small amount of calcium oxide (quick lime) in a beaker. The beaker becomes hot. $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$. Calcium hydroxide is slaked lime.
   - Whitewashing: slaked lime solution applied to walls reacts slowly with carbon dioxide in air, forming a thin layer of calcium carbonate in two to three days, which gives a shiny finish: $Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$. Marble also has the formula $CaCO_3$.
   - Since a large amount of heat is released, the reaction of quick lime with water is exothermic.
   - Other combination reactions: $C(s) + O_2(g) \rightarrow CO_2(g)$; $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$.
   - Other exothermic reactions: burning of natural gas, $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$; respiration, $C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l) + energy$; decomposition of vegetable matter into compost.

7. **Explain corrosion and rancidity as effects of oxidation in everyday life. Describe their harmful effects and how they can be prevented.** [5 marks]
   Ans. Corrosion:
   - When a metal is attacked by substances around it such as moisture and acids, it is said to corrode.
   - Examples: iron gets a reddish-brown coating (rust), silver gets a black coating and copper gets a green coating.
   - Harmful effects: it damages car bodies, bridges, iron railings, ships and all objects made of metals, especially iron. A large amount of money is spent each year to replace damaged iron.
   - Prevention: painting, oiling or greasing iron articles keeps air and moisture away.

   Rancidity:
   - When fats and oils are oxidised, they become rancid and their smell and taste change.
   - Harmful effect: food becomes unpleasant and unfit to eat.
   - Prevention: antioxidants are added to food containing fats and oils; food is kept in air-tight containers; chips packets are flushed with nitrogen, an inert gas, so that the chips do not get oxidised.